

PHYS 338K Final Project Proposal: Op-Amp Based Function Generator

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1. Project Description

This project aims to design and build a function generator capable of producing sine, square, and triangle waveforms. In response to design feedback, our implementation will avoid using any dedicated waveform generator integrated circuits (e.g. ICL8038) and will instead rely entirely on discrete operational amplifiers. This approach provides a more educational and technically engaging process by emphasizing analog circuit fundamentals.

The core of the generator will utilize a two op-amp configuration that simultaneously produces square and triangle waveforms that maintain constant amplitude as we adjust the frequency. This design choice mitigates a common limitation found in simpler integrator-based oscillators where amplitude varies with frequency. The triangle wave output will then be processed through a diode-based nonlinear waveshaping circuit to approximate a low-distortion sine wave. All waveform outputs will be buffered to ensure low output impedance and stable signal delivery to external loads.

2. Circuit Diagram

Block Diagram

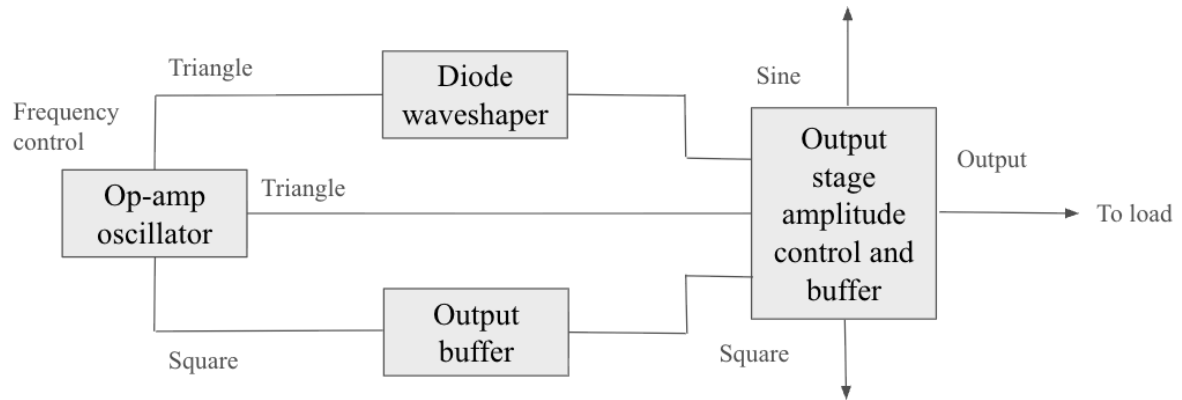


Figure 1: Function generator block diagram.

Detailed Schematic

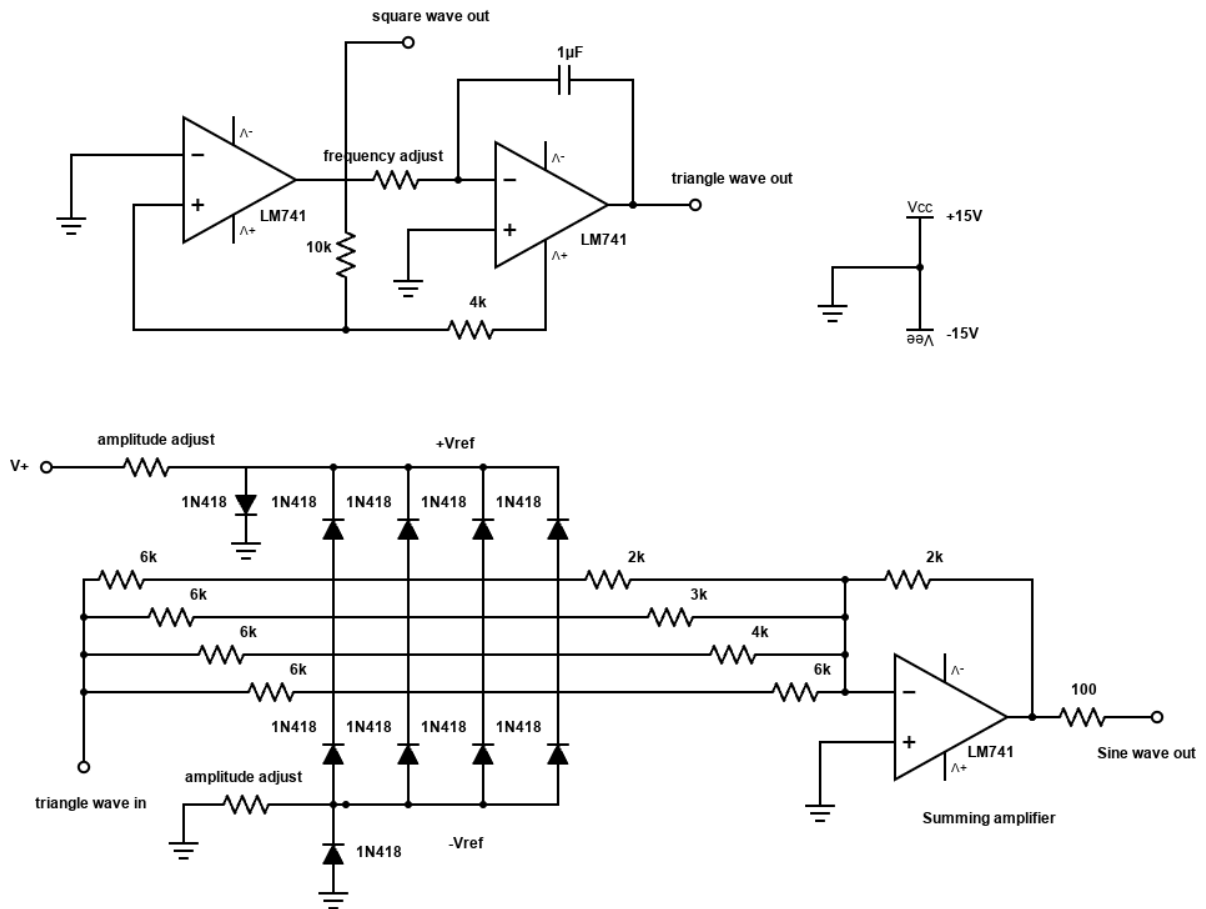


Figure 2: **Above:** A two op-amp triangle and square wave generator. The frequency of the wave can be changed by adjusting the resistance using the trimmer potentiometer without changing the amplitude. **Below:** A waveform shaping circuit to convert a triangle wave to a sine wave. The output amplitude can be changed by adjusting the resistance using the rotary potentiometers.

The wave generator circuit in Figure 2 maintains the amplitudes of the square and triangle waves constant while adjusting the frequency value. The waveform shaper circuit below it uses biased diodes to progressively clip the peaks of the triangle wave to round it to a sine wave. The resistors set the bias points for the diodes, and the op-amp sums the currents from the network.

3. Revised Parts List

- **Op-Amps:** 3 x LM741 op-amps
- **Diodes:** 10 x 1N4148 small-signal switching diodes
- **Resistors:** Multiple values for oscillator, waveshaper, and biasing (e.g., 1kΩ, 2kΩ, 4kΩ, 6kΩ, 10kΩ)

- **Capacitors:** 1 x $1\mu\text{F}$ capacitor
- **Trimmer Potentiometer:** 1 x $10\text{ k}\Omega$ for frequency control
- **Rotary Potentiometers:** 2 x $250\text{ k}\Omega$ and 2 x $500\text{ k}\Omega$ for output amplitude control
- **Switch:** 1 x SP3T (Single Pole, 3-Position) rotary switch for waveform selection
- **Power Supply:** 1 x $\pm 15\text{V}$ dual rail supply
- **Output Connector:** BNC or 3.5mm audio jack
- **Oscilloscope** 1 x Tektronix TBS1052C
- **Breadboard/PCB:** For prototyping and final assembly